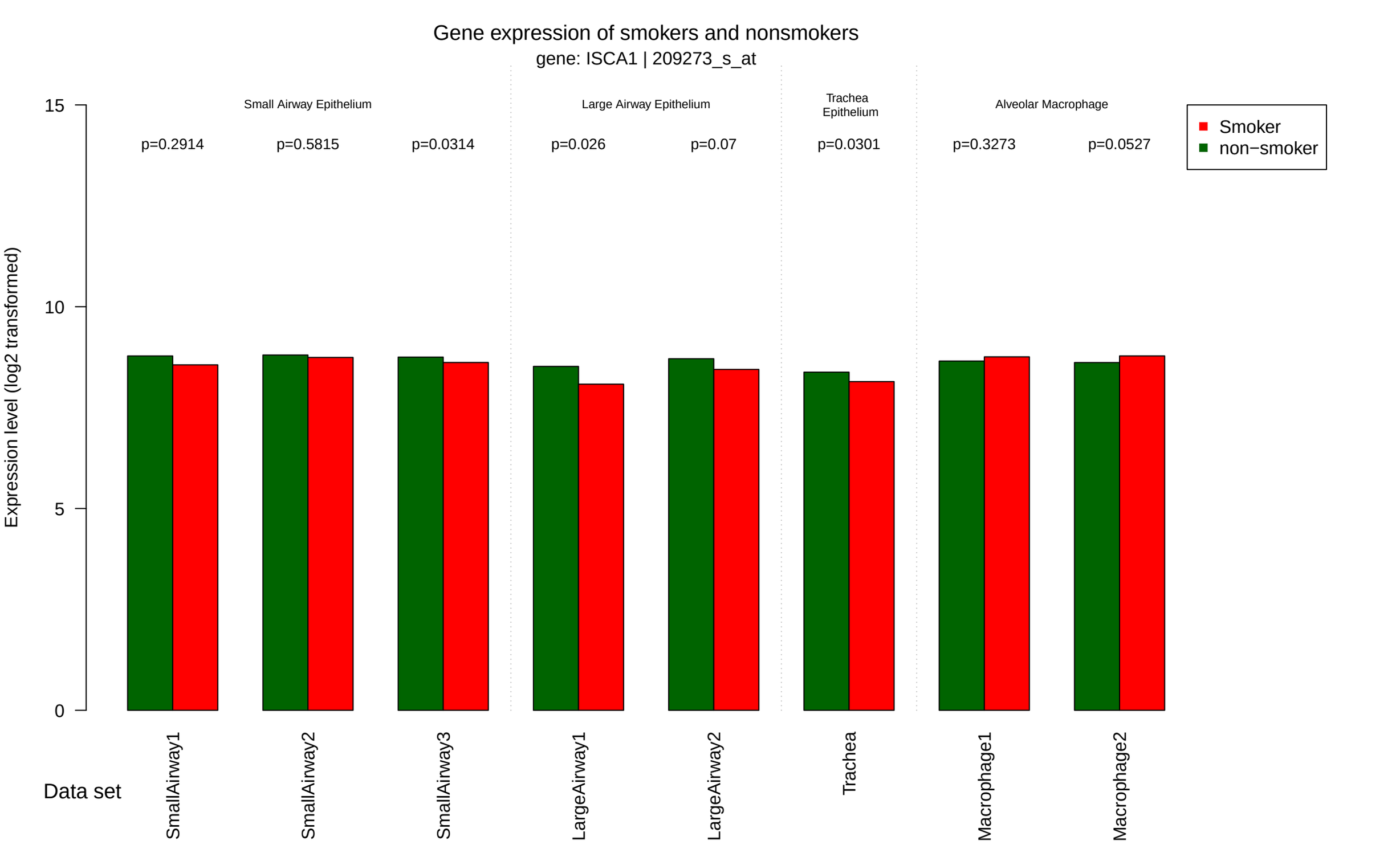
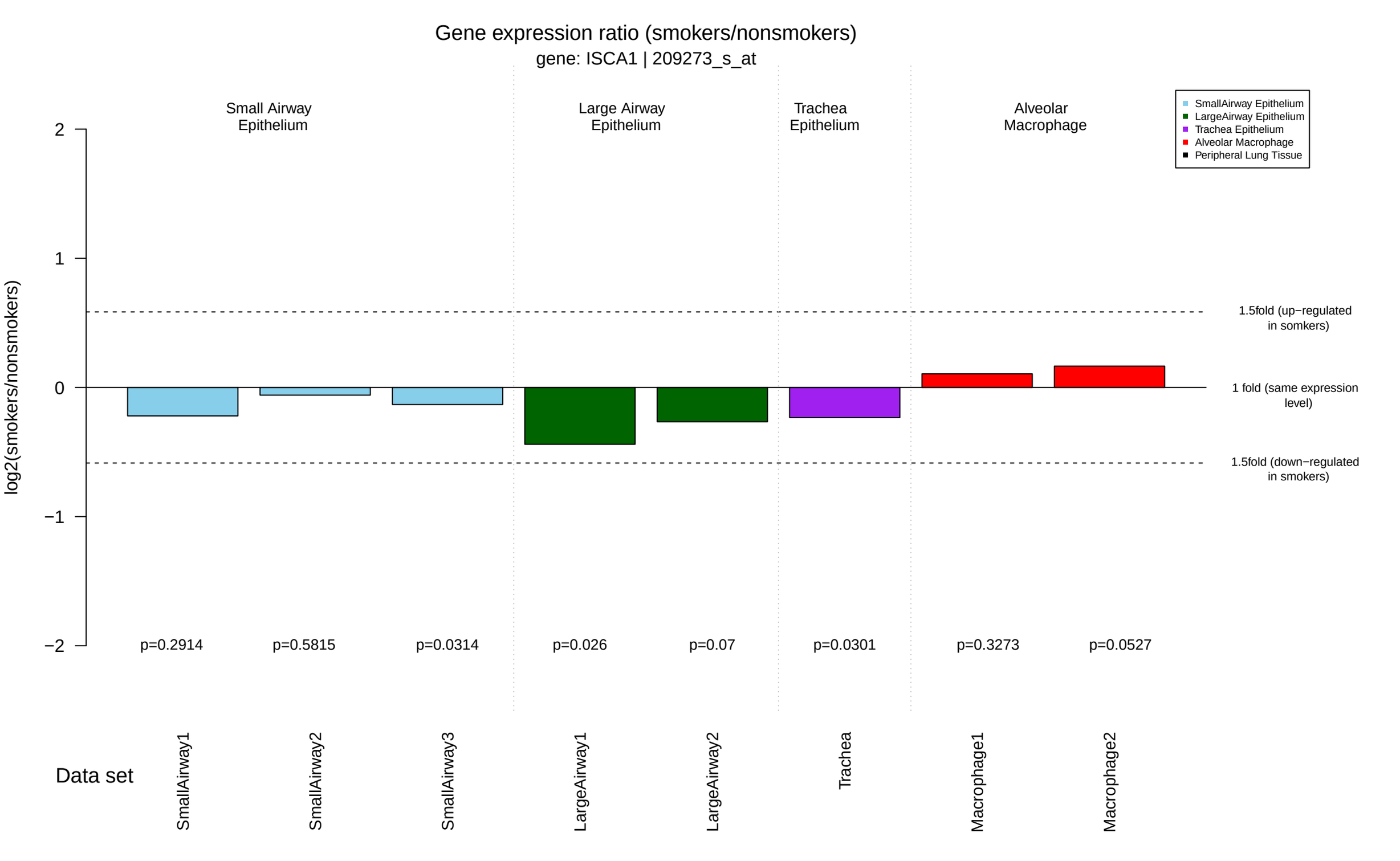
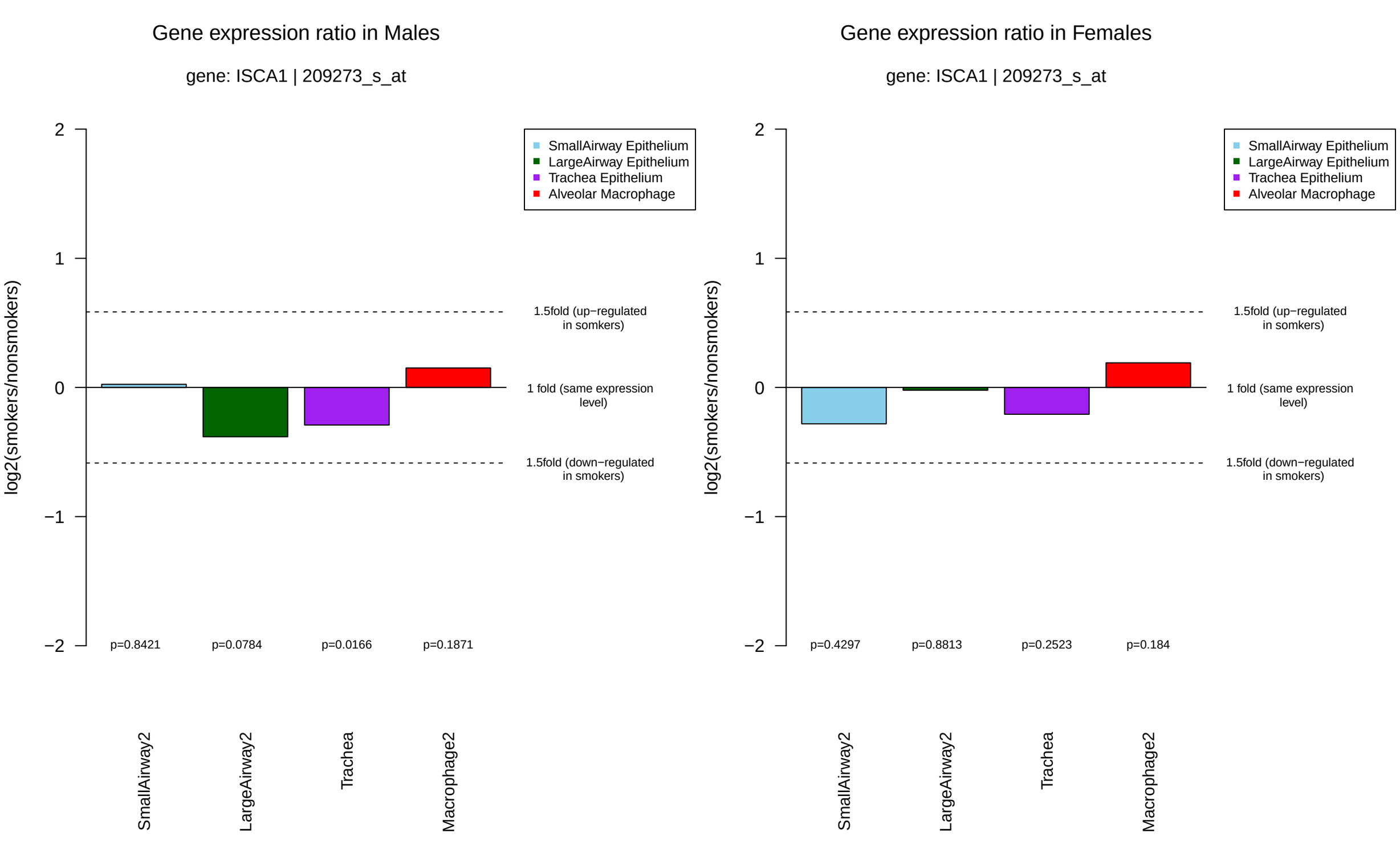
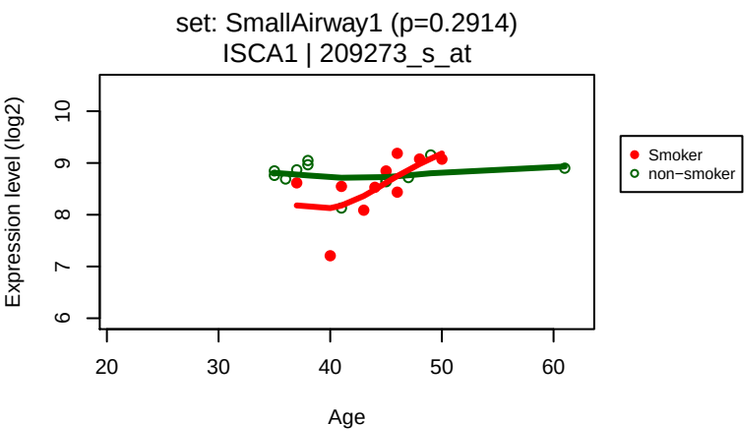
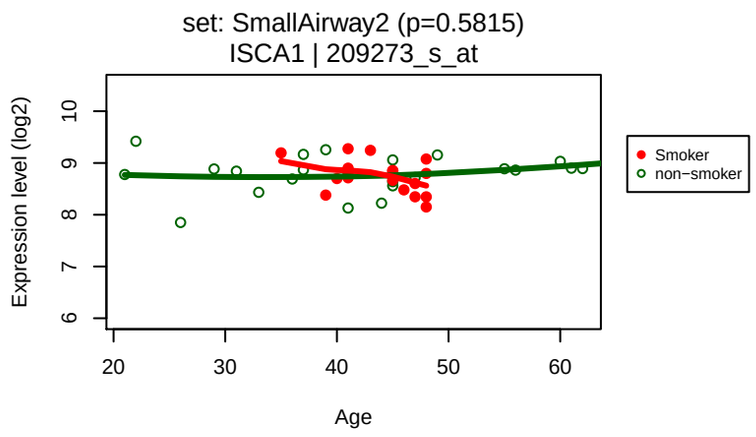
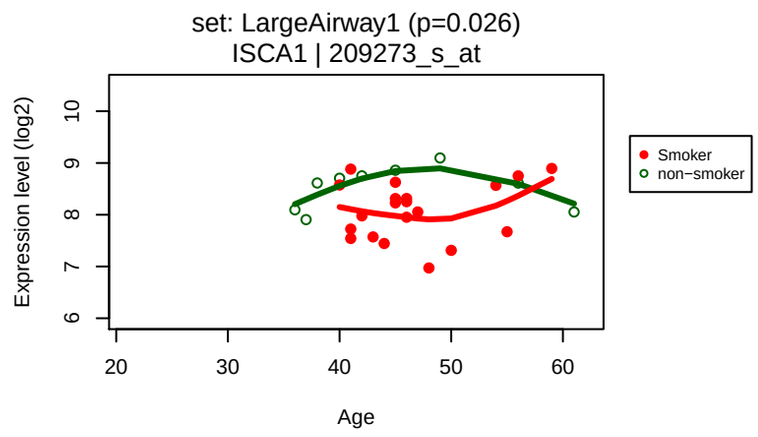

Gene expression of smokers and nonsmokers
gene: ISCA1 | 209273_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2914
p=0.5815
p=0.0314
p=0.026
p=0.07
p=0.0301
p=0.3273
p=0.0527
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

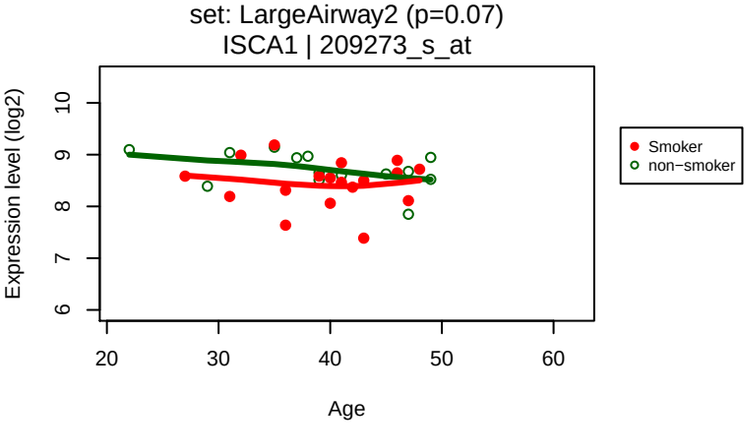

Gene expression ratio (smokers/nonsmokers)
gene: ISCA1 | 209273_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2914
p=0.5815
p=0.0314
p=0.026
p=0.07
p=0.0301
p=0.3273
p=0.0527
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

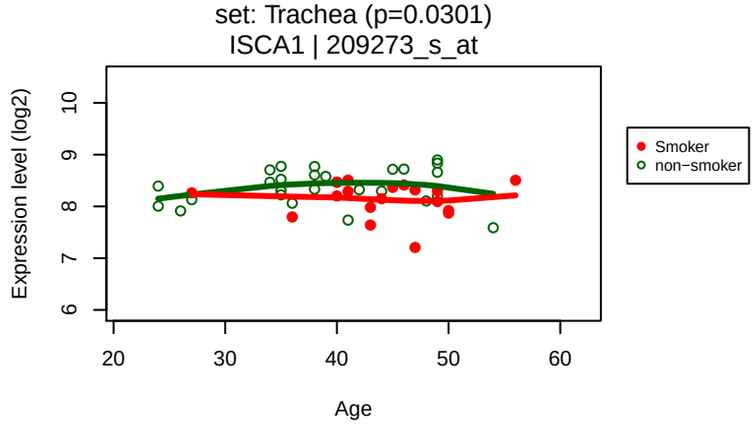

Gene expression ratio in Males
gene: ISCA1 | 209273_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8421
p=0.0784
p=0.0166
p=0.1871
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISCA1 | 209273_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4297
p=0.8813
p=0.2523
p=0.184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

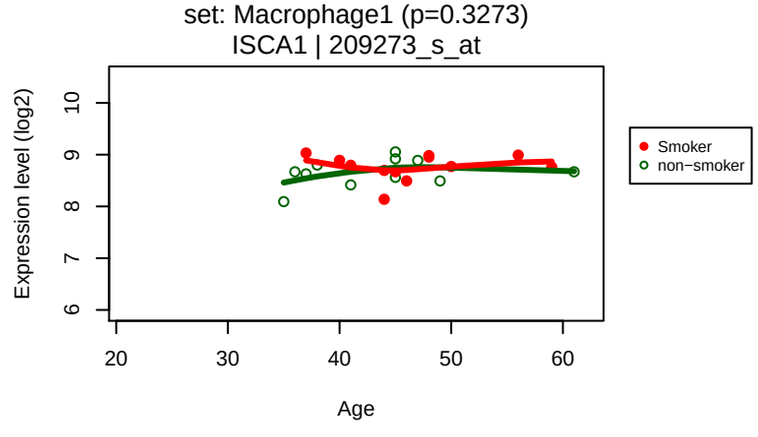

set: SmallAirway1 (p=0.2914)
ISCA1 | 209273_s_at
Expression level (log2)
Age
Smoker
non−smoker

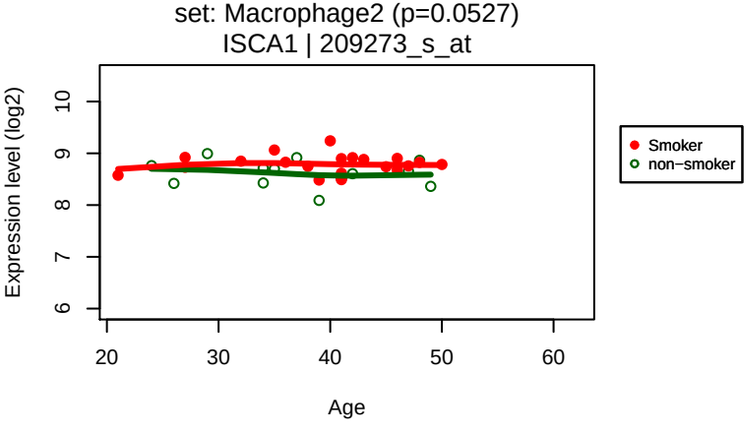

set: SmallAirway2 (p=0.5815)
ISCA1 | 209273_s_at
Expression level (log2)
Age
Smoker
non−smoker

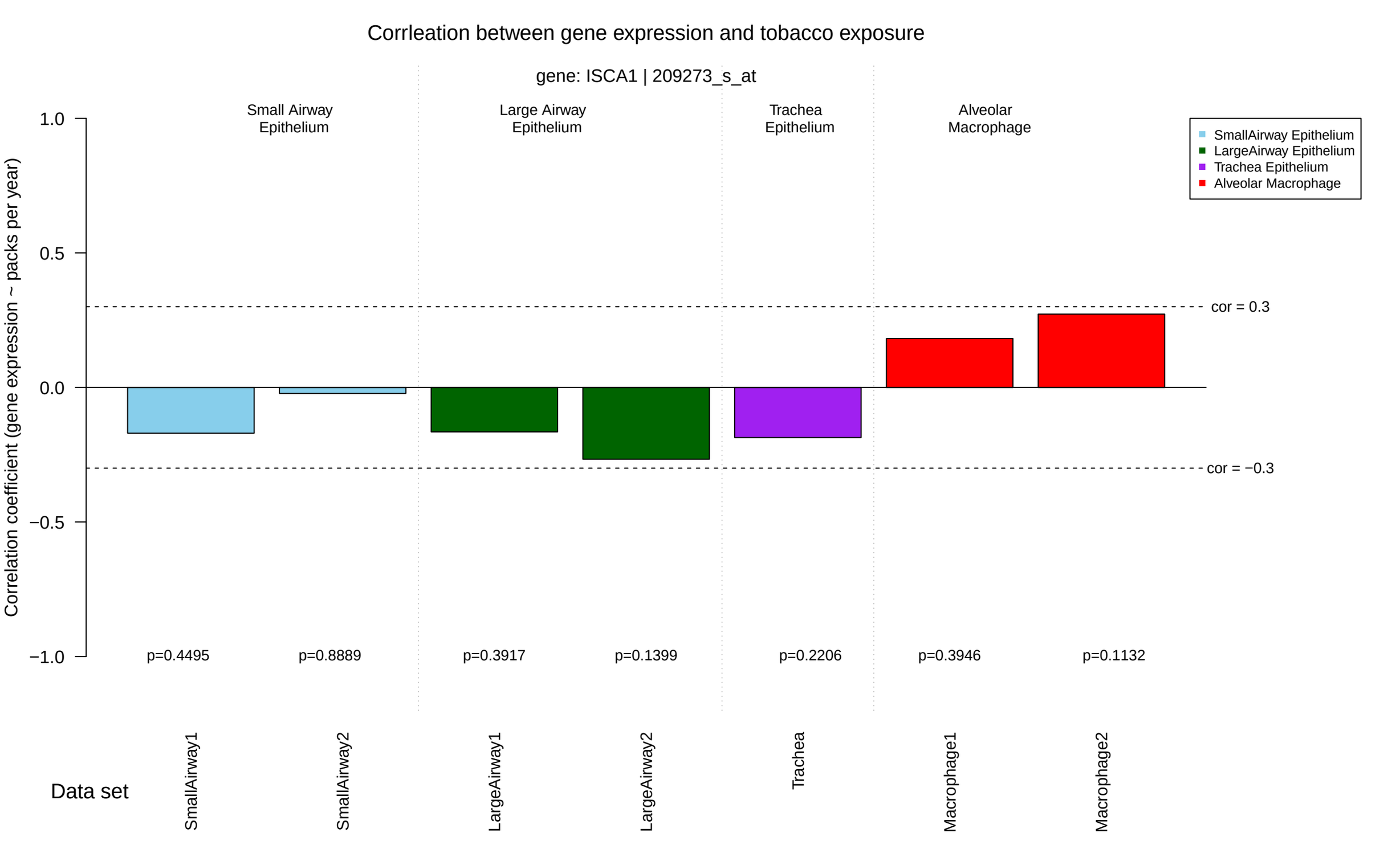

set: LargeAirway1 (p=0.026)
ISCA1 | 209273_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.07)
ISCA1 | 209273_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0301)
ISCA1 | 209273_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3273)
ISCA1 | 209273_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0527)
ISCA1 | 209273_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISCA1 | 209273_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4495
p=0.8889
p=0.3917
p=0.1399
p=0.2206
p=0.3946
p=0.1132
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2